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Cross-Asset Volatility and Equity Risk

Empirical Report

Quantitative Techniques for Financial Economics
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Preface

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1. Research Question and Economic Motivation

1.1 Research Question and Economic Motivation

Mandelbrot (1963, p. 418) observed that "large changes tend to be followed by large changes, of either sign, or small changes tend to be followed by small changes." This persistence makes a market's own history a natural starting point for volatility forecasting. It also raises a question: can information from other markets improve forecasts beyond what is already captured by past equity volatility?

Forecasts of equity volatility inform risk measurement, portfolio allocation and derivative pricing, making forecast accuracy relevant to investors and financial institutions. For researchers, examining the contribution of other markets provides a way to assess whether cross-market relationships contain useful predictive information.

Commodity and Treasury markets offer plausible sources of such information. Uncertainty in energy markets may affect expectations about production costs, inflation and corporate profitability. Gold-market volatility may reflect changing perceptions of financial risk, while Treasury yield movements may capture shifts in expectations about interest rates and economic conditions. These connections motivate investigating whether developments in other markets help anticipate future equity variance (Yongdeng Xu, et. Al., 2024).

This study therefore asks whether incorporating information from commodity and US Treasury markets improves forecasts of S&P 500 futures realised variance relative to models relying solely on historical equity information. We examine commodity futures, including crude oil, gold and natural gas, alongside a Treasury-market proxy constructed from changes in the ten-year US yield.

The central issue is the incremental forecasting value of these additional inputs. Markets may respond simultaneously to common shocks without their histories providing useful signals about future equity variance. Adding variables also increases the number of parameters to estimate, so a broader information set does not automatically produce more accurate forecasts. We therefore compare equity-only models with specifications incorporating additional market information and evaluate their out-of-sample performance. Complementary modelling approaches, including HAR and VAR models, allow us to examine whether the findings depend on how volatility dynamics are represented. Our objective is to assess predictive usefulness rather than establish causal transmission between markets (Francis X, et. Al., 2012).

2. Data

We measure volatility by realised variance, which sums squared intraday returns and therefore measures daily variance more precisely than squared daily returns, providing a reliable target for forecast evaluation (Andersen et al., 2003). We use the five-minute version (RV5), which Liu et al. (2015) find difficult to outperform across asset classes. Sampling more frequently would add market microstructure noise, while sampling less frequently would discard information.

All markets are represented by liquid futures contracts, most of which trade almost continuously on CME Group exchanges, so their variances cover comparable trading hours. For this reason, we use E-mini S&P 500 futures (ES) rather than the cash index. Crude oil is represented by the NYMEX West Texas Intermediate contract (CL), which is physically delivered in Cushing, Oklahoma. It therefore reflects US oil prices, and thus the energy costs of S&P 500 firms, more closely than the North Sea Brent benchmark, which serves mainly as a reference for European and international crude prices. Natural gas (NG) is the NYMEX Henry Hub contract, the reference price for the US gas market. Gold (GC) captures demand for a safe haven in periods of financial stress (Baur & McDermott, 2010), and corn (C), used in the HAR, adds an agricultural market not directly tied to energy and financial conditions. The ten-year Treasury yield serves as the benchmark for long-term US interest rates, which affect firms' discount rates and financing costs.

Daily RV5 and closing prices come from the Volatility Archive for Realized Estimates (VOLARE; Cipollini et al., 2026), which computes realised measures from intraday mid-quote prices supplied by Kibot. All realised variances enter in natural logarithms, which compresses their pronounced spikes. The forecasting target is equity log realised variance; the HAR additionally uses daily returns computed from closing prices. Variable definitions and descriptive statistics are reported in Tables A1 and A3.

The Treasury yield (DGS10) is published by the Board of Governors of the Federal Reserve System and obtained from FRED. As no intraday bond data are available, squared daily yield changes in basis points serve as a proxy for Treasury-market variation, although they are not directly equivalent to the intraday futures measures. Because yields are reported in whole basis points, the change is exactly zero on many days. These genuine zeros are retained, so the proxy remains unlogged.

ncomplete dates are excluded rather than filled in. Treasury changes are calculated on their original calendar and delayed by one observed Treasury date to allow conservatively for publication timing. The forecast for 13 July 2023, for example, uses a Treasury change ending no later than 11 July (Table A4).

The VOLARE data start in September 2009, but the early gold series contains missing observations and implausible extreme values, hence we start in January 2011. The common sample contains 3,910 dates from 5 January 2011 to 31 August 2026. The first 3,127 observations, up to 12 July 2023, form the training sample, and the remaining 783 the evaluation sample (Table A2 and Figure A1). We preserve this cutoff after adding Treasuries, so every specification faces the same evaluation targets and changes in sample composition cannot drive the model comparison. Corn is missing on one additional training date. We do not remove further observations, so crisis episodes such as April 2020 remain in the sample.

Source: Andersen, T. G., Bollerslev, T., Diebold, F. X., & Labys, P. (2003). Modeling and Forecasting Realized Volatility. Econometrica, 71(2), 579–625. http://www.jstor.org/stable/3082068

3. Empirical Methodology

We examine whether information from other markets improves forecasts of equity realised variance using three complementary approaches: a heterogeneous autoregressive (HAR) model, a vector autoregressive (VAR) model and a leveragerd VAR model. These approaches allow us to investigate the predictive contribution of additional market information under different model specifications. The following subsections present each approach and explain the main modelling choices.

3.1 The Heterogeneous Autoregressive (HAR) Model

To forecast the volatility of equity index futures, we employ the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) introduced by Corsi (2009). Traditional autoregressive frameworks like GARCH or short-memory ARMA specifications frequently struggle to reproduce the slow, hyperbolic decay observed in empirical financial volatility: A phenomenon commonly characterized as long memory. While fractionally integrated processes (such as ARFIMA) can replicate this persistence, they are computationally intensive and lack a structural market foundation.

The HAR framework circumvents these limitations by building upon the Heterogeneous Market Hypothesis (Müller et al., 1997). This hypothesis posits that financial markets are populated by distinct agent profiles operating across divergent temporal horizons: short-term intraday market participants and high-frequency traders who monitor and react to daily volatility shocks; medium-term institutional rebalancers and swing traders who condition their exposure on weekly market variance; and long-term fundamental asset managers and pension funds whose capital allocations depend on monthly macro volatility trends.

Rather than specifying a formal fractionally integrated model, the standard HAR process approximates long memory through a parsimonious linear aggregation of realized volatility components sampled across these three distinct frequencies: daily ($d = 1$), weekly ($w = 5$), and monthly ($m = 22$ trading days). Formally, the standard HAR model that predicts the next-day realized variance is specified as:

$$\ln RV_{t+1}^{(d)} = \beta_0 + \beta_d \ln RV_t^{(d)} + \beta_w \ln RV_t^{(w)} + \beta_m \ln RV_t^{(m)} + \epsilon_{t+1}$$

where:

$$\ln RV_t^{(w)} = \frac{1}{5}\sum_{i=0}^{4} \ln RV_{t-i}^{(d)} \quad \text{and} \quad \ln RV_t^{(m)} = \frac{1}{22}\sum_{i=0}^{21} \ln RV_{t-i}^{(d)}$$

represent the moving averages over weekly and monthly horizons.

Incorporating Asymmetries: The Leverage LHAR Model

Because negative equity returns induce a disproportionate surge in future volatility, a phenomenon attributed to financial leverage effects (Black, 1976) and volatility feedback mechanisms (Campbell & Hentschel, 1992), we augment the baseline HAR to an LHAR-RV framework following Corsi and Renò (2012). We define the daily negative return shock as:

$$r_t^- = \min(r_t, 0)$$

where $r_t = \frac{\text{ret}_t}{100}$ denotes the daily percentage returns expressed in decimals. Multi-horizon negative return cascades are constructed identically:

$$r_{t,w}^- = \frac{1}{5}\sum_{i=0}^{4} r_{t-i}^-, \quad r_{t,m}^- = \frac{1}{22}\sum_{i=0}^{21} r_{t-i}^-$$

These asymmetric downside terms are appended directly to the regressor matrix to isolate market-wide panic spillovers.

Empirical Findings

Model	R^2	Adj. R^2	AIC	BIC	MSE
Baseline HAR	0.7192	0.719	6062.97	6087.89	0.3592
HAR-X (Oil)	0.7197	0.7193	6061.91	6105.52	0.3591
HAR-X (NatGas)	0.7197	0.7192	6062.46	6106.08	0.3594
HAR-X (Gold)	0.7198	0.7194	6060.38	6103.99	0.3605
HAR-X (Corn)	0.7193	0.7188	6067.75	6111.37	0.3595
HAR-X (Energy: Oil + NG)	0.7202	0.7196	6061.00	6123.31	0.3591
HAR-X (3 Comms: CL, NG, GC)	0.7211	0.7202	6055.52	6136.52	0.3598
HAR-X (4 Comms: CL, NG, GC, C)	0.7211	0.7200	6060.92	6160.62	0.3600
Baseline LHAR	0.7456	0.7452	5681.8	5725.38	0.3213
LHAR-X (Oil)	0.7461	0.7455	5680.66	5742.91	0.3210
LHAR-X (NatGas)	0.7467	0.7460	5672.11	5734.37	0.3206
LHAR-X (Gold)	0.7463	0.7457	5676.98	5739.23	0.3225
LHAR-X (Corn)	0.7457	0.7451	5686.22	5748.48	0.3212
LHAR-X (Energy: Oil + NG)	0.7471	0.7463	5671.14	5752.07	0.3203
LHAR-X (3 Comms: CL, NG, GC)	0.7479	0.7469	5665.81	5765.42	0.3209
LHAR-X (4 Comms: CL, NG, GC, C)	0.7480	0.7468	5670.27	5788.56	0.3207

3.2 VAR

Vector autoregression

To identify the contribution of other markets, we compare VAR forecasts with an equity-only AR containing the same number of lags and calendar controls. This comparison separates cross-market information from the predictive value of equity volatility's own history. A

persistence forecast, equal to the previous retained observation, provides an additional benchmark.

The dependent variable is log RV5 of E-mini S&P 500 futures. We estimate four VAR specifications: equity with oil and gold; equity with natural gas and the Treasury proxy; equity with all four commodities, including corn; and a full system containing all six series. Each equation is estimated by OLS with an intercept, 11 month indicators and four weekday indicators. The calendar controls accommodate recurring seasonal differences without pre-filtering the data. Treasury variation enters as the supplied squared yield-change proxy. Appendix B describes the variables and common sample.

The initial training sample contains 3,127 observations through 12 July 2023. Lag selection uses only this sample: BIC selects four lags and AIC selects nine from a maximum of 20 in the full system. Four lags form the main specification; nine provide a sensitivity check. We hold the lag order fixed across models within each comparison and re-estimate on an expanding sample, producing 783 one-step forecasts from 13 July 2023 to 31 August 2026. Forecasts use earlier retained observations and assume that all inputs dated before the target are available. Appendix A reports the specification checks.

Table 1 Out-of-sample accuracy with four lags

Model	MSE	MAE	MSE difference vs AR (%)
Persistence	0.43574	0.50428	+22.16
AR	0.35669	0.45396	0.00
VAR: oil and gold	0.35786	0.45575	+0.33
VAR: gas and treasury	0.35838	0.45651	+0.48
VAR: all commodities	0.35897	0.45759	+0.64
Full VAR	0.35918	0.45785	+0.70

Notes: 783 common forecast targets; errors are measured in equity log RV5. All AR/VAR models include identical calendar controls. Source: own calculations using the group dataset.

The equity AR reduces MSE by 18.14% relative to persistence. None of the four-lag VARs improves on the matched AR in either MSE or MAE. The full VAR's MSE is 0.70% higher, while adding oil and gold raises it by 0.33%. The improvement over persistence therefore comes from modelling equity's own dynamics and calendar effects; it cannot be attributed to the additional markets.

The nine-lag oil-and-gold VAR achieves a small 0.14% MSE reduction relative to AR(9), but its MAE is higher. Broader systems again fail to improve MSE. First differencing and an additional Treasury delay also leave the AR–VAR comparison unchanged (Appendix A). Mixed stationarity evidence and remaining residual autocorrelation warrant caution with conventional Granger-test inference. Overall, additional markets provide little incremental forecast value for this equity benchmark over the selected evaluation period. This finding does not establish that spillovers are absent or that the same result holds for other equity markets or forecast horizons.

3.3 Jake´s model LVAR + Restricted LVAR

4. Empirical results

5. Economic interpretation

The empirical results suggest that incorporating information from commodity and Treasury markets provides limited additional economic value when forecasting S&P 500 futures volatility. The equity-only AR model achieves a mean squared error (MSE) of 0.356682, compared with 0.435742 for the persistence benchmark, representing an improvement of approximately 18.14%. In contrast, adding oil and gold increases the MSE by 0.33%, while including all four additional markets increases it by 0.66%. These relatively small differences indicate that incorporating cross-market information does not meaningfully improve forecasting accuracy.

From an economic perspective, these findings suggest that historical equity volatility already captures much of the information relevant for predicting future market uncertainty. Although commodity and equity markets are interconnected through common macroeconomic shocks, inflation expectations, and changes in investor sentiment, these relationships do not necessarily translate into additional predictive value.

The findings also have practical implications for financial institutions and investors, who rely on volatility forecasts for risk management, portfolio allocation, and Value-at-Risk calculations. Given the limited improvements in forecasting accuracy, the additional complexity and data requirements of multivariate models may outweigh their practical benefits. Consequently, simpler models relying on historical equity volatility may offer a more efficient forecasting approach.

Overall, the results highlight the economic importance of volatility persistence while suggesting that incorporating additional market information provides limited incremental forecasting value within the specifications and evaluation period examined.

Consistent with Corsi (2009), who finds that a simple autoregressive model of realized volatility forecasts S&P 500 futures volatility well, the equity market's own history appears to capture most of the relevant information. Two mechanisms may explain why the linkages between markets do not translate into forecasting gains. First, common shocks are likely to show up in equity variance on the same day, so lagged commodity or bond information adds little to lagged equity variance. Second, transmission may run mainly from equities to other markets. Diebold and Yilmaz (2012) find that volatility spillovers across US stock, bond, foreign exchange and commodity markets were limited before the 2007 crisis. As the crisis intensified, the spillovers ran particularly from the stock market to the other markets. Predictive gains from outside the stock market also tend to be episodic. Paye (2012) finds that forecasts based on macroeconomic variables struggle to outperform a univariate benchmark out-of-sample, with predictive power concentrated around the onset of recessions. Our evaluation period (July 2023 to August 2026) contains no recession or crisis comparable to 2008 or 2020, so such episodes have little opportunity to appear.

Banks and other financial institutions use volatility forecasts, for example, to calculate Value-at-Risk, which rises and falls with the forecast volatility. A forecast that changes MSE by less than 1% would therefore hardly change the resulting Value-at-Risk, whereas modelling the equity market's own history improves accuracy by 18% relative to the persistence benchmark. Such accuracy matters most at short horizons like the one-day horizon studied here. Christoffersen and Diebold (2000) show that volatility forecastability decays quickly with the horizon and is mainly relevant for short-horizon risk management. We evaluate statistical accuracy only, however, and do not test whether the forecasts lead to better risk-management decisions.

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6. Robustness and limitations

Several limitations qualify the interpretation of our findings. First, Treasury-market variation is measured using squared daily changes in the ten-year US yield rather than intraday realised variance. Although bond prices and yields are closely linked, this daily proxy captures net yield movements and misses fluctuations that reverse within the day. It therefore differs from the RV5 measures used for equity and commodity futures. Our Treasury results concern the predictive value of this particular proxy and do not establish whether Treasury futures realised variance would provide additional forecasting information.

Second, our findings relate specifically to S&P 500 futures and should not be generalised to equity markets as a whole. This is a liquid and widely followed market, and the index's sector composition affects its exposure to developments in commodity markets. Oil-market information could have greater predictive value for indices with larger energy-sector weights or for markets in economies more dependent on oil production or consumption. Less liquid equity markets may also incorporate information at a different speed, potentially changing the usefulness of lagged cross-market signals. These possibilities would require a separate analysis.

Third, the results depend on the sample period, forecast horizon and model specifications considered. Relationships between markets may change with economic conditions, so limited average forecasting gains do not rule out useful contributions during particular episodes. Similarly, information that adds little to the specifications examined here could prove useful at other horizons or in models that allow relationships to change over time. Our conclusions therefore concern the approaches evaluated rather than every possible use of cross-market information.

Finally, the VAR diagnostics indicate remaining limitations in the model specification. The stationarity tests provide mixed evidence, while residual autocorrelation and changing error variance suggest that the models do not fully capture the underlying dynamics. These findings warrant caution when interpreting coefficient tests and small differences in forecast accuracy. The observed forecast rankings should not be treated as conclusive evidence of general model superiority. The VAR analysis also does not evaluate whether the forecasts produce meaningful improvements in portfolio allocation or risk-management decisions.

7. Conclusion

8. Division of labour

Task	*Lead*	*Supporting contributor*
Research question and economic motivation	**Amina**	**Femke**
Data	**Femke**	**Amina**
HAR model	**Dean**	**Amina**
VAR model	**Tim**	**Femke**
LVAR model	**Jake**	**Dean**
Empirical results	**All team members**	**—**
Economic interpretation	**Femke**	**Amina**
Robustness and limitations	**Tim**	**—**

5. Economic interpretation

5.1 Magnitude and Economic Significance of the Findings

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Empirical appendix

Chapter 1

Xu, Y., Guan, B., Lu, W., & Heravi, S. (2024). Macroeconomic shocks and volatility spillovers between stock, bond, gold and crude oil markets. *Energy Economics*, *136*, 107750.

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